

Lecture 2.

Let $M = R$ -module, $m \in M$, $Rm = \langle m \rangle = \{ rm \mid r \in R \}$

$$\text{Ann}_R(m) = \{ r \in R \mid rm = 0 \} \quad \text{annihilator}$$

check. this is an ideal.

Then we have $R/\text{Ann}_R(m) \rightarrow Rm$ is an isomorphism.

All cyclic module Rm isomorphic to some quotient ring R/I , where $I = (\text{Ann}_R(m))$ is an ideal.

Def. An element $m \in M$ is called

- a torsion element if $\text{Ann}_R(m) \neq 0$
- a torsion free - element if $\text{Ann}_R(m) = 0$.

Def: $\text{tor}(M) = \{ m \in M \mid \text{Ann}_R(m) \neq 0 \}$

lem: if R is an ID, the $\text{tor}(M)$ is an R -submod.

Proof: $m \in \text{tor}(M)$, $r \in R$, $rm \in \text{tor}(M)$

$$m, n \in \text{tor}(M), \exists \lambda, u \in R, \lambda \neq 0, u \neq 0,$$

$$\lambda m = 0 = un.$$

$$(\lambda u)(m+n) = 0.$$

Ex: ① $0 \in M$ is torsion.

② R comm. ring. R is ID iff ${}_R R$ is torsion free.

or we can write $\text{tor}({}_R R) = 0$. In this case $\text{tor}(R^n) = 0$.

③ If $I =$ non-zero ideal, $\text{tor}(R/I) = R/I$.
 R -module.

$\nexists R/I$ as R/I -mod. (R , comm.).

$\nexists$ is torsion free $\nexists I$ is prime.

④. more generally. I : non-zero ideal M : R -mod.

$$\text{tor}(\underbrace{M/IM}_{R\text{-mod}}) = M/IM.$$

Prop. R , comm. M : R -mod $r \in R$.

$f: M \rightarrow M$ is a morphism of R -modules.
 $x \mapsto rx$

If moreover M is torsion-free. $r \neq 0$.

$M \rightarrow M$ is a monomorphism.
 $x \mapsto rx$

Lemma: If R : ID. M : R -module. then.

(1). $\text{tor}(\text{tor}(M)) = \text{tor}(M)$.

(2). $\text{tor}(M/\text{tor}(M)) = 0$.

Proof (1). check

(2). Assume. $m \in M$, $r \in R$ $r(m + \text{tor}(M)) = 0$.

$$\Leftrightarrow rm \in \text{tor}(M), \exists \overset{\neq 0}{r'} \in R \text{ s.t. } r'rm = 0.$$

Since $r' \neq 0$ hence $m \in \text{tor}(M)$

$$\Rightarrow \text{tor}(M/\text{tor}(M)) = 0.$$

Let J : set, $R^J = \prod_{j \in J} R$ $R^{(J)} = \bigoplus_{j \in J} R$.

Let. M : R -mod. $x_j, j \in J$ a family of elements of M .

Then the submodule generated by $(x_j)_{j \in J}$ is denoted by

$$\sum_{j \in J} R x_j.$$

We have a morphism of R -modules.

$$\varphi: R^{(J)} \longrightarrow M.$$

$$(r_j)_{j \in J} \longmapsto \sum_{j \in J} r_j x_j.$$

- φ is injective $\Leftrightarrow \{x_j\}$ are linearly independent.
- φ is surjective $\Leftrightarrow \{x_j\}$ generate M .
- φ is isomorphism $\Leftrightarrow M$ is free with basis $\{x_j\}$.

Lemma: Let L be a free R -mod with basis $\{x_i\}_{i \in I}$.

For any R -mod M the following map

$$\text{Hom}_R(L, M) \longrightarrow M^I.$$

$$\phi \longmapsto (\phi(x_i))_{i \in I} \text{ is a bijection.}$$

Prop: Let L be a free R -mod.

(1). Let $\pi: X \rightarrow Y$ be an epimorphism of R -mod. For any

R -mod homomorphism $\varphi: L \rightarrow Y$ there exists a morphism.

$\tilde{\varphi}: L \rightarrow X$ s.t. the diagram ~~(free $\Rightarrow$ proj.)~~

$$\begin{array}{ccc} & \varphi_L & \\ \tilde{\varphi} \swarrow & \downarrow & \\ X & \xrightarrow{\pi} & Y \longrightarrow 0 \end{array} \quad \text{commute.}$$

(2). In particular, for any R -mod M , any epimorphism π

$\pi: M \rightarrow L$ has a section, that is, there is a morphism

$\sigma: L \rightarrow M$ with $\pi\sigma = \text{id}_L$. In this case, σ is inj.

$\text{Im } \sigma \simeq L$ and $M = \ker \pi \oplus \text{Im } \sigma$.

$$0 \rightarrow \ker \pi \rightarrow M \xrightarrow{\pi} L \rightarrow 0$$

$\sigma \uparrow$

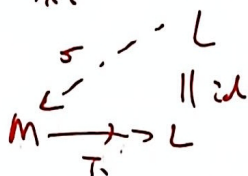
(M is R -mod. N is R -submodule. N is called a direct summand of M if there is a submodule K s.t. $M = N \oplus K$)

proof: (1) let $\{e_i\}_{i \in I}$ be a basis of L .

Let $\{x_i\}_{i \in I} \subset X$ s.t. $\pi(x_i) = \varphi(e_i)$

define $\tilde{\varphi}(e_i) = x_i$

(2) as "1."



This property is the property of proj. module. this shows that all free modules are proj. mods. $\Rightarrow$ the P is proj.

Exercise. L : free R -mod. P : a direct summand of L .

show whenever $X \xrightarrow{\pi} Y$ is an epi. and $\varphi: P \rightarrow Y$ is a morphism. there exists a morphism $\tilde{\varphi}: P \rightarrow X$ s.t. $\varphi = \pi \circ \tilde{\varphi}$

Tensor product.

设 M, N 是 R 上的模, $f: M \times N \rightarrow K$ 是双线性映射, 则存在唯一的线性映射 $\tilde{f}: M \otimes_R N \rightarrow K$ 使得 $\tilde{f}(m \otimes n) = f(m, n)$.

R : comm. ring. M, N, K : R -mod. tensor prod.

$M \otimes_R N$ "linearize the bilinear map"

$f: M \times N \rightarrow K$ is bilinear if.

$$f(x+x', y) = f(x, y) + f(x', y)$$

$$f(x, y+y') = f(x, y) + f(x, y')$$

$$f(ax, y) = a f(x, y) = f(x, ay)$$

for all $a \in R$

$x, x' \in M$

$y, y' \in N$

$$f(x, 0) = 0 = f(0, y)$$

eg. $f(-x, y) = f(x, -y) = -f(x, y)$

^{the}
~~Def~~ Tensor product of M, N is a pair.

(3)

$(M \otimes_R N, T_{M,N})$ where $M \otimes_R N$ is an R -module and

$T_{M,N} : M \times N \rightarrow M \otimes_R N$ is a bilinear map.

usually denote by $T_{M,N}(m,n) = m \otimes n$ which satisfies the following universal properties:

whenever X is an R -module and $f : M \times N \rightarrow X$ is a bilinear map there exists a unique R -module morphism.

$\tilde{f} : M \otimes_R N \rightarrow X$ s.t. the diagram commutes.

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & X \\ T_{M,N} \downarrow & \nearrow \tilde{f} & \\ M \otimes_R N & & \end{array}$$

$\Leftarrow$ up-to iso. ~~the~~ tensor is ~~unique~~ unique.

Suppose. (T_1, t_1) (T_2, t_2) two tensor products.

$$\begin{array}{ccc} M \times N & \xrightarrow{t_1} & T_1 \\ t_2 \downarrow & \nearrow f & \\ T_2 & \xleftarrow{g} & \end{array}$$

$$t_2 = g t_1, \quad t_1 = f t_2$$

$$\Rightarrow t_2 = g f t_2$$

Since g, f are unique

$$\Rightarrow g f = \text{id}_{T_2} \quad f g = \text{id}_{T_1}$$

$$\Rightarrow T_2 \cong T_1$$

Existence : here F be a free R -mod with basis $M \times N$

let B be the submodule of F gene. by

$$\ell(m_1 + m_2, n) = \ell(m_1, n) + \ell(m_2, n)$$

$$\ell(m, n_1 + n_2) = \ell(m, n_1) + \ell(m, n_2)$$

$$\lambda \ell(m, n) = \ell(\lambda m, n) \quad \lambda \ell(m, n) = \ell(m, \lambda n)$$

$$m, m_1, m_2 \in M, \quad n, n_1, n_2 \in N, \quad \lambda \in R$$

Define $M \otimes_R N = F/B$. $T_{M,N} : M \times N \rightarrow M \otimes_R N$
 $(m, n) \mapsto \ell(m, n) + B$

(i). check: the map $T_{M,N}$ is well-defined and bilinear.

(ii). check: the pair $(M \otimes_R N, T_{M,N})$ is tensor product

The set of elements $\{m \otimes n \mid m \in M, n \in N\}$ called elementary tensors. generates $M \otimes_R N$

Fix $m \in M$, the map $N \rightarrow M \otimes_R N$ is a morphism. of
 $n \mapsto m \otimes n$

R -mod. Its image $m \otimes_R N := \{m \otimes n \mid n \in N\}$ is submodule of $M \otimes_R N$, isomorphic to a quotient of N

prop: 11. we have isomorphisms.

$$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$$

$$(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$$

And, $M_1 \otimes_R M_2 \rightarrow M_1 \otimes_R M_1$

$$m_1 \otimes m_2 \mapsto m_2 \otimes m_1$$

12. As well as $M_1 \otimes (M_2 \oplus M_3) \rightarrow (M_1 \otimes M_2) \oplus (M_1 \otimes M_3)$

$$m_1 \otimes (m_2 + m_3) \mapsto m_1 \otimes m_2 + m_1 \otimes m_3$$

And more general $M \otimes_R (\bigoplus_{i \in I} M_i) \xrightarrow{\sim} \bigoplus_{i \in I} (M \otimes_R M_i)$

$$m \otimes (\sum_{i \in I} m_i) \mapsto \sum_{i \in I} m \otimes m_i$$

13. as well as: $R \otimes_R M \xrightarrow{\sim} M$ and more general.

$$r \otimes m \mapsto rm$$

$$R^{(I)} \otimes_R M \xrightarrow{\sim} M^{(I)}$$

$$(\lambda_i)_{i \in I} \otimes m \mapsto (\lambda_i m)_{i \in I}$$

proof: $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$

$$(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$$

Let $m_3 \in M_3$ fixed. define $\phi_{m_3}: M_1 \times M_2 \rightarrow M_1 \otimes (M_2 \otimes M_3)$
 $(m_1, m_2) \mapsto m_1 \otimes (m_2 \otimes m_3)$

Then $\exists!$ morphism $\psi_{m_3}: M_1 \otimes M_2 \rightarrow M_1 \otimes (M_2 \otimes M_3)$

Next define $(M_1 \otimes M_2) \times M_3 \longrightarrow M_1 \otimes (M_2 \otimes M_3)$

$$\left(\sum_{i \in I} \lambda_i m_i^{(1)} \otimes m_i^{(2)}, m_3 \right) \mapsto \sum \lambda_i m_i^{(1)} \otimes (m_i^{(2)} \otimes m_3)$$

check bilinearity.

$\Rightarrow \exists !$ R -mod home. $\tilde{f} : (M_1 \otimes M_2) \otimes M_3 \longrightarrow M_1 \otimes (M_2 \otimes M_3)$
 $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$

Similarly, $M_1 \otimes (M_2 \otimes M_3) \longrightarrow (M_1 \otimes M_2) \otimes M_3$
 $m_1 \otimes (m_2 \otimes m_3) \mapsto (m_1 \otimes m_2) \otimes m_3$ is an R -isomorphism.

Then M, N, P R -module, we have a canonical iso.

$$\phi_{M,N,P} : \text{Hom}_R(M \otimes_R N, P) \longrightarrow \text{Hom}_R(M, \text{Hom}_R(N, P))$$

$$f \mapsto \phi(f) : M \longrightarrow \text{Hom}_R(N, P)$$

$$m \mapsto \phi(f)(m) : N \longrightarrow P$$

$$n \mapsto \phi(f)(m)(n)$$

$$= f(m \otimes n)$$

$$\left(\begin{array}{ccc} \psi(g) & & \\ \uparrow & \nearrow & \downarrow \\ M \otimes N & \xrightarrow{\psi(g)} & P \\ \uparrow & & \uparrow \\ M \times N & & (m,n) \end{array} \right) \xleftarrow{\psi} g$$

if R comm. this is R -mod. iso.

not is $\mathbb{Z}$ -mod. iso.

"Canonical"

$$\begin{array}{ccc} \text{Hom}_R(M \otimes_R N, P) & \xrightarrow{\phi_{M,N,P}} & \text{Hom}_R(M, \text{Hom}_R(N, P)) \\ \uparrow \text{Hom}_R(f \otimes \text{id}, P) & & \uparrow \text{Hom}_R(f, \text{Hom}_R(N, P)) \\ M & & \\ f \downarrow & & \\ M' & & \text{Hom}_R(M' \otimes_R N, P) \xrightarrow{\phi_{M',N,P}} \text{Hom}_R(M', \text{Hom}_R(N, P)) \end{array}$$

explain: $M \otimes N \xrightarrow{f \otimes \text{id}} M' \otimes N \xrightarrow{\tau} P$
 $\downarrow f \otimes \text{id} \quad \downarrow$
 $M' \otimes N \quad f(M \otimes N)$

Consider. $M \xrightarrow{f} M' \quad N \xrightarrow{g} N'$

$$f \otimes g : M \otimes N \rightarrow M' \otimes N'$$

$$m \otimes n \mapsto f(m) \otimes g(n).$$

Definition: M, N R -modules.

$L^n(M, \dots, M_n; N) = \{f \mid f: M_1 \times \dots \times M_n \rightarrow N \text{ is } n\text{-linear}\}$

$$L^2(M, N; P) \cong L(M \otimes_R N; P) = \text{Hom}_R(M \otimes_R N, P)$$

$$L(M \otimes_R N; P) \cong L(M, L(N, P))$$

Prop: Let M, N be f, g free R -mod. denote.

$$M^\vee = \text{Hom}_R(M, R)$$

Then $M^\vee \otimes_R N \cong \text{Hom}_R(M, N)$ is an isomorphism.

Proof: $M^\vee \times N \rightarrow \text{Hom}_R(M, N)$

$$(f, n) \mapsto \varphi(f, n) : M \rightarrow N$$

$$m \mapsto f(m)n.$$

This is bilinear. hence φ induces an R -mod-homo-

$$\tilde{\varphi} : M^\vee \otimes_R N \rightarrow \text{Hom}_R(M, N)$$

$$f \otimes n \mapsto \varphi(f, n).$$

On the other hand. Let $v_1, \dots, v_n$ be a basis of M .

define dual basis $v_i^\vee \in M^\vee : v_i^\vee(v_j) = \delta_{ij}$

Define $\text{Hom}_R(M, N) \rightarrow M^\vee \otimes_R N$

$$g \mapsto \sum_{i=1}^n v_i^\vee \otimes g(v_i).$$

Check. $\varphi \tilde{\varphi} = \text{id}_{\text{Hom}_R(M, N)}$ $\tilde{\varphi} \varphi = \text{id}_{M^\vee \otimes_R N}$

prop: ①. Assume there R is a subring of R' contain 1.

②

Let M be an R -mod. Then the bilinear map.

$$R' \times (R' \otimes_R M) \rightarrow R' \otimes_R M$$

$$(\lambda', \lambda' \otimes m) \mapsto \lambda' \lambda' \otimes m$$

defines a structure of R' -module on the R -module $R' \otimes_R M$.

More generally, let $f: R \rightarrow T$ be ring homomorphism.

View T as an R -mod defined by $R \times T \rightarrow T$
 $(\lambda, u) \mapsto f(\lambda)u$.

The bilinear map $T \times (T \otimes_R M) \rightarrow T \otimes_R M$

$$(u, u \otimes m) \mapsto u u \otimes m$$

defines a T -mod structure on the R -module $T \otimes_R M$.

Exercise: Let I be an ideal of R . M an R -module

show the morphism of R -mods

$$M \rightarrow (R/I) \otimes_R M$$

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism $M/IM \cong (R/I) \otimes_R M$ as R/I -mod

Let R be a ring (not nec. comm.) M be a right R -mod. M_R
 N be a left R -mod. ${}_R N$

Define. a balanced product of M and N to be a pair.

(P, f) where P is an abelian group, and f is a map

$M \times N \rightarrow P$ satisfied.

$$(1) f(x+x', y) = f(x, y) + f(x', y)$$

$$(2) f(x, y+y') = f(x, y) + f(x, y')$$

$$(3) \lambda f(x, y) = f(\lambda x, y) = f(x, \lambda y)$$

$$\forall x, x' \in M \\ y, y' \in N \\ a \in R$$

- An ~~tensor~~ ^{of M and N} prod. of M and N is a balanced product
 $(M \otimes_R N, \tau_{M \otimes N})$ s.t. (φ, f) is any balanced product
 of M and N , then exists a unique group homo.
 abelian.

$f: M \otimes_R N \rightarrow P$ s.t. the following diagram commutes

